Structural Consistency Is Not Retrieval Utility: An Exact-and-Heuristic Audit of Preference-Graph Repair for Multi-Ranker Retrieval

Soroush Vahidi[1*]

[1*]Ying Wu College of Computing, New Jersey Institute of Technology, Newark, NJ, USA.

Corresponding author(s). E-mail(s): sv96@njit.edu;

Abstract

Purpose: This study asks whether enforcing acyclicity in derived multi-ranker preference graphs, by heuristic or exact minimum-weight feedback-arc-set repair, produces a statistically reliable improvement in retrieval effectiveness, or whether structural consistency and retrieval utility must be treated as separate quality dimensions.

Methods: Stored BM25, TF-IDF, and MiniLM scores were converted into query-specific weighted preference graphs on four public retrieval benchmarks under three vote-construction regimes. Rankings were extracted from unrepaired, greedy-repaired, and exactly repaired graphs, using paired query-level inference with Holm correction. Exact repair was solved to certified optimality with the open-source SCIP solver as a methodological control on heuristic suboptimality.

Results: Graph construction strongly determined cyclicity: conservative graphs were acyclic in the observed evidence, whereas the permissive regime produced cyclic graphs for most queries. Repair removed non-trivial contradictory edge weight and, when the candidate pool exceeded the evaluation cutoff, changed evaluated top-ranked membership. However, no repaired-versus-unrepaired normalized discounted cumulative gain comparison survived Holm correction in the canonical, larger-pool, or exact-repair comparison families.

Conclusion: In this evidence, preference-graph repair is a real structural intervention but not a validated surrogate for improving retrieval effectiveness. Repair claims should therefore report structural diagnostics and downstream retrieval metrics separately.

Keywords: preference graphs, feedback arc set, minimum-weight feedback arc set, graph repair, rank aggregation, information retrieval, retrieval evaluation, data quality

1 Introduction

Pairwise preferences are a natural and increasingly common form of ranking evidence. Rather than committing to a single scalar relevance score, a pairwise judgment states only that one document should be ranked ahead of another for a given query. This form of evidence arises whenever several ranking signals are available for the same query – lexical and dense retrievers, cross-encoders, or large language models used as pointwise, pairwise, or listwise judges – and it arises directly whenever a system elicits pairwise comparisons instead of absolute scores. Aggregating such judgments across several rankers or several repeated comparisons for the same query produces a natural representation: a query-specific weighted directed preference graph, in which an edge $u \to v$ records support for ranking document u ahead of document v, and the edge weight records how strongly that support should count.

That representation is attractive because it preserves disagreement rather than collapsing it. It is also structurally fragile. Because the graph is built from several partially disagreeing sources, it can contain directed cycles – $u \to v \to w \to u$ – so that no single total order is consistent with every retained edge. Cycles of this kind are not a pathological edge case; they are the expected outcome whenever comparison evidence is aggregated across imperfectly correlated sources. Because a cycle is a case where the evidence contradicts itself, it is natural to treat acyclicity, or a reduction in cyclic and contradictory structure, as a desirable property of the graph, and cycle repair (removing or reversing a minimum-weight set of conflicting edges) as an attractive preprocessing step before a ranking is read off the graph. This intuition is plausible on its face: a graph that no longer contradicts itself seems like a strictly better input to whatever ranking rule is applied to it next.

That plausibility, however, rests on treating four distinct stages of a preference-graph pipeline as if they were one. *Construction* decides which pairwise votes are retained and how they are weighted, and is itself sensitive to score normalization, vote semantics, and thresholding choices made before any graph exists. *Structural repair* operates on the resulting graph and optimizes a purely graph-internal objective – the total weight of edges that must be removed or reversed to restore acyclicity. *Ranking extraction* reads a final ordering off the (possibly repaired) graph using some scoring rule. *Downstream evaluation* then compares that ordering against relevance judgments using a retrieval metric that has no direct relationship to the repair objective. Repair minimizes disagreement with the graph's own edges; retrieval metrics score agreement with an external notion of relevance. Nothing in the construction of the repair objective guarantees that reducing the former improves the latter, and nothing in the standard presentation of repair work makes that gap explicit.

This gap is the question prior work leaves open: does improving a preference graph's structural consistency – as measured by the repair objective itself – also improve the retrieval effectiveness of the ranking eventually read off that graph? Existing evidence does not answer this cleanly, for three reasons that compound. First, essentially all empirical evaluations of preference-graph repair use heuristic repair methods, so any observed retrieval effect, positive or null, is open to the objection that it reflects the heuristic's suboptimality rather than a property of repair itself: a null result could always be dismissed as "the heuristic simply did not repair enough."

Second, graph construction is rarely held explicit and varied alongside repair, so it is difficult to attribute an observed effect (or its absence) specifically to the repair step rather than to upstream choices that determine how much contradictory structure the graph contained in the first place. Third, studies that report both structural diagnostics (cyclicity, removed edge weight) and a retrieval metric do not generally test whether the two move together under corrected statistical inference; an improvement in the former is often reported alongside, and implicitly read as evidence for, the latter.

This paper addresses that gap directly. We hold graph construction explicit across three vote-construction regimes that vary how much contradictory structure a graph retains, and we compare, on the same underlying candidate sets and ranker scores, *unrepaired*, *greedy-repaired*, and *exactly repaired* preference graphs. The greedy-only design that dominates prior work cannot, by itself, resolve the central question: a null retrieval result under a heuristic repair method is always consistent with the hypothesis that a better repair would have revealed a gain the heuristic missed. We therefore include exact minimum-weight feedback-arc-set (MWFAS) repair, solved to certified optimality with an open-source mixed-integer solver, as a diagnostic control rather than as a proposed practical method: if exact repair, which by construction cannot be out-performed by any heuristic on the repair objective itself, still fails to produce a reliable retrieval gain, then heuristic suboptimality is ruled out as the explanation for a null result. This is what exact repair is for in this study – not a faster or better production repair algorithm, but a diagnostic instrument that isolates whether the retrieval question has an answer the heuristic could plausibly have obscured.

The paper's scope is stated honestly rather than maximally. The main empirical setting is score-derived multi-ranker retrieval: preference graphs built from stored BM25, TF-IDF, and MiniLM scores across four public benchmarks spanning distinct retrieval domains. A smaller real-large-language-model pairwise-judgment pilot is included only as a bounded, directional addendum, not as a second confirmatory study; it is not treated as validating or generalizing the main findings. The paper does not claim state-of-the-art retrieval effectiveness, does not propose a new minimum-weight feedback-arc-set formulation or solver, does not claim that repair never helps retrieval in any setting, does not claim that exact repair universally fails to help, and does not offer a validated criterion for deciding, per query, when repair is worth applying.

Within that scope, the central finding is restrained rather than sweeping. Structural repair is genuinely active: under the permissive construction regime, both greedy and exact repair remove a measurable share of total graph weight, and evaluating against a candidate pool larger than the metric cutoff shows that repair can change which documents even enter the evaluated top-k set. What does not follow is a reliable retrieval benefit. Across four benchmarks, three construction regimes, and both repair methods, no repaired-versus-unrepaired normalized discounted cumulative gain (nDCG) comparison survives Holm-corrected paired significance testing. The paper's thesis is therefore that structural graph consistency and downstream retrieval utility are related but empirically distinct quality dimensions, and that enforcing acyclicity should not be treated as a validated surrogate for improving retrieval effectiveness without evidence obtained under the retrieval metric itself.

This paper is a rigorous empirical and computational study of that question, not a proposal for a new or superior reranking method. Its contributions are:

1. A controlled empirical framework that separates preference-graph construction, structural repair, ranking extraction, and downstream retrieval evaluation into explicit, independently varied stages, so that any observed effect can be attributed to a specific stage rather than to their combination.
2. A three-way comparison, under identical construction and evaluation protocols, of unrepaired, greedy-repaired, and exactly (SCIP-based MWFAS) repaired preference graphs – using exact repair as a diagnostic control that closes the standard objection that a null retrieval result merely reflects heuristic under-repair.
3. A four-benchmark, multiplicity-corrected empirical finding: graph repair is structurally active and can change which documents are evaluated at all, yet no repaired-versus-unrepaired nDCG family – canonical, larger-pool, or exact-repair – survives Holm-corrected paired testing, so structural consistency is not shown to be a reliable general surrogate for retrieval utility.
4. An open, reproducible implementation and evidence pipeline covering graph construction, greedy and exact repair, ranking extraction, and the statistical evaluation protocol, so that the audit reported here can be rerun, checked, or extended by others.

2 Related Work

2.1 Pairwise Ranking and Preference Aggregation

Representing evidence as pairwise comparisons rather than absolute scores is classical in ranking theory. Given comparisons $u \succ v$, two largely separate traditions convert them into a ranking. The first is *probabilistic preference modeling*: Bradley–Terry-type models explain observed comparison outcomes through latent per-item strengths estimated by maximum likelihood [4, 20], and Plackett–Luce models generalize the same idea to full or partial orderings [30, 37]. Rank Centrality recovers a ranking as the stationary distribution of a Markov chain built from pairwise comparisons, without fitting a parametric strength model [33], and combinatorial Hodge theory decomposes a pairwise comparison graph into a gradient (globally consistent) component and a curl/harmonic (locally or globally cyclic) residual, giving a spectral account of how far the data are from admitting a single ranking [23, 49]. Bias-aware pairwise ranking further shows that even the sampling process that generates which pairs are compared can itself introduce systematic ranking bias, independent of any repair step [15].

The second tradition is *combinatorial rank aggregation and cycle repair*, which treats a set of pairwise preferences (or several ranked lists) as combinatorial input and asks for an ordering, or a modification of the input, that minimizes a specific disagreement objective. Kemeny's optimal-consensus ranking minimizes total pairwise disagreement with a set of input rankings [25]; web-scale rank aggregation work popularized Markov-chain, Copeland-style (wins-minus-losses), and Borda-style (positional) aggregation rules and analyzed their properties [10]; and follow-up work gave approximation algorithms and complexity results for Kemeny-optimal and related aggregation problems, connecting them to correlation clustering [1, 26]. Rank comparison metrics that tolerate ties and partial overlap, such as the Fagin–Kumar family, are a companion line of work for measuring how much two rankings (or a ranking before and after a repair step) actually differ [13, 14].

These two traditions answer different questions. Probabilistic models estimate a ranking that best explains noisy comparison outcomes and treat residual disagreement as statistical noise around a latent order. Combinatorial repair instead treats the observed graph as fixed evidence and asks how to modify it, at minimum cost, so that a single consistent order becomes possible. The present paper is positioned entirely in the second tradition: it uses combinatorial minimum-weight feedback-arc-set repair (Section 3.4) on an already-constructed preference graph, and its research question is not about estimating latent item strengths but about whether the combinatorial repair objective, once optimized, is diagnostic of a different objective – retrieval effectiveness – that it was never designed to optimize.

2.2 Pairwise and Graph-Based Ranking in Information Retrieval

Pairwise comparison has become a common building block of modern retrieval and reranking pipelines. Pairwise ranking prompting elicits document-level comparisons directly from a large language model and aggregates them into a ranking [39], and pointwise, pairwise, and listwise LLM reranking paradigms have been compared directly for effectiveness and cost [45]. Listwise and setwise rerankers reduce the number of language-model calls needed to reorder a candidate list [17, 38, 55, 59], and hybrid rerankers combine lexical, dense, and cross-encoder signals within a single scoring or fusion step [29, 34, 51, 57]. Graph-free score and rank fusion – reciprocal rank fusion, CombSUM, and Borda-style combination – remains a strong and widely used alternative to any graph-based aggregation step, in part because it avoids comparing incompatible raw score magnitudes [2, 5, 8, 16, 28, 32, 50].

A smaller body of work builds an explicit preference graph from pairwise document judgments before extracting a ranking, rather than fusing scores or ranks directly. `LLM-RankFusion` aggregates several LLM rankers by explicitly modeling and mitigating pairwise inconsistency in the resulting comparisons [56], and confidence-aware or bias-aware pairwise aggregation methods weight or correct individual comparisons before they are combined [15]. This is the setting the present paper studies: a query-specific weighted directed graph built from several ranking signals, from which a ranking is extracted before or after a structural repair step. Standard retrieval benchmarking practice – reusing stored candidate pools and score files rather than re-running upstream retrieval, and evaluating with rank-sensitive metrics such as nDCG – follows the same evaluation conventions established for zero-shot retrieval benchmarking more generally [9, 46].

2.3 Inconsistency, Cycles, and Preference Repair

A pairwise preference graph is transitive, and therefore admits a single consistent total order, only in the special case that it is acyclic. In general it is not: aggregating several partially disagreeing sources routinely produces directed cycles, including simple two-way contradictions and longer cyclic chains. The combinatorial problem of removing the least-weight set of edges needed to make a weighted directed graph acyclic is the (minimum-weight) feedback arc set problem. It is NP-hard in general [24]; classical work gives fast, deterministic greedy heuristics that repeatedly remove the lightest edge on a discovered cycle [11], and later work gives polynomial-time approximation schemes and connects the problem to correlation clustering [1, 26]. This paper does not claim the minimum-weight feedback-arc-set formulation itself as a contribution – it is a long-studied classical combinatorial optimization problem – and instead studies the empirical consequence of solving it, both heuristically and to certified optimality, for a downstream retrieval task.

The same structural issue has recently reappeared in the context of large language models used as pairwise or comparative judges: repeated or multi-evaluator LLM comparisons frequently produce transitivity violations and directed cycles, which several recent studies diagnose or attempt to resolve [19, 43, 58]. That work generally treats acyclicity (or reduced contradiction) as a desirable property of the resulting evaluation and reports downstream gains on tasks such as model ranking, response selection, or fine-tuning-data selection [19]; it does not test whether the acyclicity objective itself is diagnostic of retrieval effectiveness under a corrected, paired statistical protocol, which is the specific gap this paper addresses.

This paper substantially extends and revises an earlier preprint by the same author [48], which studied a smaller and partially overlapping evidence base under a less conservative evaluation protocol (no separation of canonical from larger-pool evaluation cells, no Holm correction applied to the headline retrieval claim, and no exact-repair comparison) and reported a conditional, dataset-dependent positive retrieval effect. That preprint has not completed peer review at its original target venue and is cited here only as a preprint, not as a prior peer-reviewed publication. The present study is not a resubmission of it: it uses a materially more conservative statistical protocol, adds exact SCIP-based repair specifically to rule out heuristic-repair artifacts as an explanation, and reaches a different, more restrained conclusion on an updated evidence base. Readers should treat the two as distinct, non-comparable analyses rather than as an earlier and later version of the same result.

2.4 Structural Quality Versus Downstream Utility

A recurring pattern across the work reviewed above is that structural or intrinsic quality – cyclicity, removed feedback-arc weight, agreement with a probabilistic comparison model, or acyclicity of an aggregated evaluation graph – is measured and reported, but rarely tested directly against an independent downstream task metric under corrected statistical inference. This mirrors a broader distinction in the data-quality literature between *intrinsic* quality (properties of the data or artifact itself) and *contextual* or task-relative quality (fitness for a specific use) [52]; work on data cascades and benchmark label error shows concretely that upstream curation or construction choices can shape downstream conclusions in ways that are easy to miss when only the final task metric is inspected [35, 42]. Simple significance testing is also known to be an unreliable guide to retrieval system comparisons unless multiplicity and pairing are handled correctly [6, 21, 47], and the interpretation of a non-significant result requires care distinct from ordinary significance testing [27].

Few studies in the preference-graph or feedback-arc-set literature isolate structural improvement from downstream task utility as two separately measured and separately tested quantities. Most either report only structural diagnostics, or report a task metric without a repair-method control (e.g. exact repair) that would rule out heuristic suboptimality as the explanation for a null or weak downstream effect. This paper's contribution relative to this literature is exactly that controlled separation: it holds graph construction explicit, compares unrepaired, heuristically repaired, and exactly repaired graphs on the same evidence, and tests the structural-repair-to-retrieval-utility link directly under a multiplicity-corrected paired protocol, rather than assuming that a gain on one objective implies a gain on the other.

3 Background and Problem Formulation

This section fixes notation and formalizes the four stages distinguished in Section 1 – retrieval setting, derived preference graph, structural inconsistency, and repair – before turning to ranking extraction and evaluation. Every definition below is stated at the level of generality needed to state the paper's research question precisely; the specific benchmarks, ranker families, and evaluated cells that instantiate this formulation are given in Section 4.

3.1 Retrieval and Candidate Rankings

Let $q \in \mathcal{Q}$ denote a query drawn from a fixed query set, and let $D_q = \{d_1, \ldots, d_n\}$ denote its query-specific candidate document set: the pool of documents that will be compared and ranked for q. Candidate sets are formed upstream of this study's pipeline, typically as the union of documents retrieved or scored by one or more base rankers, and are held fixed once formed.

A base ranker (or, more generally, a judgment source) $s \in S$ assigns a score $x_{q,s}(d) \in \mathbb{R}$ to each document $d \in D_q$ it scores; a ranker need not score every document in D_q, so scores may be missing for some (q, s, d) triples. Because different rankers' raw scores are not generally comparable in scale or distribution, scores are normalized per query and per ranker before they are compared across rankers. The normalization used throughout the canonical experiments is per-query, per-ranker min-max scaling,

$$\bar{x}_{q,s}(d) = \frac{x_{q,s}(d) - \min_{d' \in D_q} x_{q,s}(d')}{\max_{d' \in D_q} x_{q,s}(d') - \min_{d' \in D_q} x_{q,s}(d')}, \tag{1}$$

which maps every ranker's retained scores for a given query into a common $[0, 1]$ range. Section 3.2 builds pairwise votes from $\bar{x}_{q,s}$ rather than from $x_{q,s}$ directly; the raw, unnormalized construction is retained only as an explicit robustness ablation, not as an alternative primary protocol, precisely because raw heterogeneous score margins are not a scale-neutral starting point for comparing rankers.

Relevance judgments (qrels) associate each query-document pair with a graded or binary relevance label, drawn from each benchmark's original human-annotated judgments; they are external to, and never derived from, any ranker score or preference graph. All retrieval metrics reported in this paper are computed against these external judgments, evaluated at a fixed prefix length k (the evaluation cutoff): a metric such as nDCG@k scores only the top-k documents of a candidate ranking. Because the candidate pool D_q evaluated by a ranking method may be larger than the cutoff k, the relationship between the pool size $P = |D_q|$ (or a truncated evaluation pool size) and k is itself an explicit design choice returned to in Section 4: repair can only change which documents occupy the top-k set when $P > k$ gives it room to do so.

3.2 Derived Weighted Preference Graph

For each query q, the normalized ranker scores $\bar{x}_{q,s}$ are converted into a weighted directed preference graph

$$G_q = (V_q, E_q, w_q), \qquad V_q = D_q, \tag{2}$$

in which a directed edge $(u, v) \in E_q$ records retained evidence that document u should be ranked ahead of document v, and the edge weight $w_q(u, v) > 0$ records the strength of that retained evidence. The graph is a *derived* data object, not ground truth: it is the output of an explicit construction procedure applied to the normalized scores, and its structure – which edges exist, in which direction, and with what weight – depends on that procedure's choices, not on any property of the documents themselves.

Construction proceeds ranker by ranker and pair by pair. For each ranker s and each unordered document pair $\{u, v\} \subseteq D_q$ that s scores, s casts a directional vote for whichever of $u \succ v$ or $v \succ u$ its normalized scores support, with vote strength derived from the normalized score margin $|\bar{x}_{q,s}(u) - \bar{x}_{q,s}(v)|$. A pair with an exactly equal normalized score, or for which s does not score both documents, abstains rather than casting an imputed vote: ties and missing comparisons are treated as an absence of evidence, not as evidence of indifference. Votes across rankers are then aggregated per directed pair, and a pair-level and graph-level retention rule (the *vote-construction regime*) decides which directions are actually retained as edges of G_q.

This paper studies three explicit vote-construction regimes that hold the candidate set and underlying ranker scores fixed and vary only the retention rule:

- `ms2` (conservative): retain a direction only when it has majority-style support across rankers and the aggregated margin exceeds a fixed threshold. This suppresses most direct two-way contradictions and, in the canonical package, produces graphs that are nearly always acyclic.
- `ms1` (permissive): retain every ranker-supported directional vote. Because different rankers may disagree, both (u, v) and (v, u) can be retained for the same unordered pair, producing direct mutual contradictions and, transitively, longer cycles.
- `ms1_drop_mutual`: start from `ms1` and remove every unordered pair that would otherwise be retained in both directions, before aggregation. This isolates how much of `ms1`'s cyclicity is attributable specifically to direct mutual contradiction rather than to longer-cycle nontransitivity.

These three regimes are not alternative proposals for a "correct" construction rule; they are a controlled sweep whose purpose is to make graph construction, rather than the repair algorithm, a first-class experimental factor (Section 4 reports how strongly this factor determines observed cyclicity). Because a conflicting vote-construction regime can produce a graph in which repair has little or no structural work to do, an observed repair effect (or its absence) cannot be interpreted without knowing which regime produced the graph it was applied to.

3.3 Cycles and Structural Inconsistency

A weighted directed graph G_q is *acyclic* if it contains no directed cycle $u_1 \to u_2 \to \cdots \to u_m \to u_1$; equivalently, its strongly connected components (maximal vertex sets within which every vertex is reachable from every other) are all singletons. G_q admits at least one linear extension consistent with every retained edge direction if and only if it is acyclic; a cyclic G_q cannot be represented exactly by any single total order over D_q; some retained edge must be violated by any linear ranking of the graph's vertices.

This paper reports two purpose-specific structural inconsistency measures, rather than a single omnibus cyclicity score. The first is a per-query binary indicator of whether G_q is cyclic at all under a given construction regime, aggregated across queries into a cyclic-query rate. The second decomposes that rate into cyclicity attributable to direct mutual contradiction (a pair retained in both directions) versus cyclicity that survives after mutual pairs are removed (comparing `ms1` to `ms1_drop_mutual`), because these two sources of cyclicity have different plausible causes – near-tied evidence between two rankers versus genuine longer-range nontransitivity among three or more documents – and conflating them would over- or under-state how much of the observed cyclicity a repair step is actually resolving.

Acyclicity is a purely structural property: it guarantees that G_q admits *some* consistent total order, but it does not by itself guarantee that the retained edges are correct, that the graph's implied ranking agrees with relevance judgments, or that any particular extraction rule applied to an acyclic graph will outperform one applied to a cyclic graph. Transitivity, acyclicity, evidential correctness, and retrieval optimality are four distinct properties; this paper treats none of them as substitutes for any other, and Sections 3.4 and 3.5 keep the objective that repair optimizes (feedback-arc weight) explicitly separate from the metric evaluation uses (nDCG against qrels).

3.4 Minimum-Weight Feedback-Arc-Set Repair

Given a (possibly cyclic) preference graph $G_q = (V_q, E_q, w_q)$, repair seeks a minimum-weight set of edges $F_q^\star \subseteq E_q$ whose removal leaves an acyclic graph:

$$F_q^\star = \arg\min_{F \subseteq E_q} \sum_{e \in F} w_q(e) \quad \text{s.t.} \quad (V_q,\, E_q \setminus F) \text{ is acyclic.} \tag{3}$$

This is the minimum-weight feedback arc set (MWFAS) problem (Section 2.3); the repaired graph is $\widetilde{G}_q = (V_q, E_q \setminus F_q^\star, w_q)$. The objective in Equation (3) is entirely graph-internal: it minimizes disagreement with G_q's own retained edges, subject to the feasibility condition that the result be acyclic. It contains no reference to relevance judgments, ranking metrics, or any notion of retrieval utility – a property that is central to this paper's research question rather than an incidental modeling detail.

Two solution methods for Equation (3) are used and reported as co-equal, not as a primary method with a secondary check. **Greedy repair** is a fast, deterministic heuristic: while $\widetilde{G}_q$ contains a directed cycle, find one such cycle and remove its minimum-weight edge, repeating until the graph is acyclic. This does not generally reach the optimum of Equation (3), but it is cheap enough to run at the scale of the full query set. **Exact repair** solves Equation (3) to certified optimality via a standard linear-ordering mixed-integer program: for every ordered pair of nodes (u, v), a binary variable $\text{before}[u, v]$ indicates that u precedes v in a chosen total order, subject to antisymmetry ($\text{before}[u, v] + \text{before}[v, u] = 1$) and transitivity constraints over every ordered triple of nodes; the objective minimizes the total weight of edges whose direction disagrees with the chosen order,

$$\min \sum_{(u,v)\in E_q} w_q(u, v)\, \text{before}[v, u], \tag{4}$$

which is exactly Equation (3) restricted to orderings representable as a linear extension. This program is solved with the open-source SCIP solver; a query graph is included in the exact-repair evidence only when the solver certifies proven optimality, not merely a feasible incumbent.

Exact repair is used here as a *diagnostic control*, not as a proposed practical replacement for greedy repair: it does not scale to the largest query graphs, and this paper makes no claim about the general scalability or practicality of exact MWFAS solving. Its role is narrower and specific to this study's question. Because $F_q^\star$ from exact repair is, by construction, a global optimum of Equation (3), no repair method can remove less weight while still restoring acyclicity. If exact repair's downstream retrieval effect is still statistically indistinguishable from greedy repair's, then the suboptimality of the greedy heuristic is ruled out as an explanation for a null or weak retrieval effect – precisely the objection a greedy-only study cannot itself close (Section 1).

3.5 Ranking Extraction and Evaluation

A final query-specific ranking is extracted from either the unrepaired graph G_q or the repaired graph $\widetilde{G}_q$, or from the upstream ranker signal alone. The canonical experiments compare four families of extraction rule, evaluated in both unrepaired and repaired form where the rule is graph-dependent:

- **Prior-only**: rank by the upstream per-document score (e.g. a normalized score sum across rankers), independent of any graph.
- **Graph-free fusion**: reciprocal rank fusion, CombSUM, and Borda-style fusion over the constituent rankers' individual rankings, which likewise does not construct or consult a preference graph.

- **Graph-only extraction**: a Copeland-style score $s_{\text{cop}}(u) = \deg^+(u) - \deg^-(u)$ (unweighted out-degree minus in-degree); a weighted balance score $s_{\text{bal}}(u) = \sum_{(u,v)\in E} w(u,v) - \sum_{(v,u)\in E} w(v,u)$ (weighted out-edge mass minus in-edge mass); and one or more stationary-distribution scores computed from a Markov chain over the (weighted) graph, in the spirit of Rank Centrality (Section 2.1). Section 4 gives the precise, distinct instantiations this study evaluates (a damped chain and an undamped Rank Centrality chain are both used, as two related but separately reported methods, not interchangeable names for one method) together with the further graph-centrality and probabilistic-comparison extraction rules (PageRank, Bradley–Terry, topological orderings) included in specific comparisons. Each graph-only rule is computed on both G_q (unrepaired) and $\widetilde{G}_q$ (repaired) wherever both are defined, holding the extraction rule fixed so that any difference is attributable to repair rather than to a change of scoring rule.
- **Hybrid prior-plus-graph extraction**: the graph score is combined with the normalized prior score,

$$h_q(d) = (1-\alpha)\,\hat{s}_q^{\text{prior}}(d) + \alpha\,\hat{s}_q^{\text{graph}}(d), \tag{5}$$

  with both components normalized query-wise and $\alpha \in [0,1]$ fixed in the primary analysis (with a sweep reported as a robustness check). When $\hat{s}_q^{\text{graph}}$ is computed from G_q the method is unrepaired; when computed from $\widetilde{G}_q$ it is repaired.

Retrieval effectiveness is measured with normalized discounted cumulative gain (nDCG), the standard graded-relevance ranking metric [22]: for a ranking with relevance labels $\text{rel}_1, \ldots, \text{rel}_k$ at ranks $1, \ldots, k$,

$$\text{DCG@}k = \sum_{i=1}^{k} \frac{2^{\text{rel}_i} - 1}{\log_2(i+1)}, \qquad \text{nDCG@}k = \frac{\text{DCG@}k}{\text{IDCG@}k}, \tag{6}$$

where IDCG@k is the DCG of the ideal (qrels-sorted) ranking at the same cutoff, so nDCG@$k \in [0,1]$ with higher values indicating better agreement with the relevance judgments. nDCG is computed per query and compared between the unrepaired and repaired variant of the same extraction rule, holding every other choice fixed; Section 4 specifies the paired, multiplicity-corrected inference protocol used to test whether an observed repaired-versus-unrepaired difference is reliable.

The formulation above makes explicit why improving Equation (3) is not, by construction, the same as improving nDCG. The repair objective is computed entirely from G_q's own edges and weights, which are themselves a derived function of upstream normalization, vote semantics, and thresholding choices (Section 3.2); nDCG is computed from an extraction rule's output ranking against relevance judgments that never enter the repair objective at all. A smaller feedback-arc weight, or proven-optimal removal of that weight, is evidence about internal consistency of the retained votes – it is not, on its own, evidence about agreement with relevance judgments. Whether the two are empirically linked in this study's evidence, and how reliably, is the question Sections 4 and 5 take up directly.

4 Methodology

Section 3 formalized the objects this study reasons about – candidate rankings, the derived preference graph, structural inconsistency, MWFAS repair, and ranking extraction – at the level of generality needed to state the paper's research question precisely. This section instantiates that formulation: the specific research questions it answers, the concrete pipeline, datasets, rankers, construction protocol, repair implementations, extraction methods, baselines, metrics, and statistical-inference procedure used to produce the evidence reported in Section 5. No outcome values are reported here; this section documents what was measured and how, not what was found.

4.1 Research Questions

The study answers four questions, each mapped to one of the paper's contributions (Section 1):

RQ1 How much structural inconsistency – cyclicity and contradictory edge mass – is present in derived multi-ranker preference graphs, and how strongly does it depend on graph-construction choices (Section 3.2) rather than on the underlying documents?
RQ2 How do greedy and exact minimum-weight feedback-arc-set repair (Section 3.4) differ in the structural change they produce – removed edge weight and the chosen feedback arc set itself – on the same graphs?
RQ3 Does preference-graph repair, heuristic or exact, produce a statistically reliable improvement in downstream retrieval effectiveness (nDCG) relative to the unrepaired graph, once paired inference and multiplicity correction are applied?
RQ4 Are the answers to RQ1–RQ3 robust to graph-construction regime, candidate-pool policy, extraction rule, and evaluation-cutoff choices – and, specifically, is any observed retrieval null in RQ3 attributable to greedy-repair suboptimality rather than to repair itself?

These four questions are the only ones this study is designed to answer. Consistent with the scope stated in Section 1, no research question here concerns learned policies for choosing when or how to invoke repair in a deployed system; that adjacent research thread uses an unrelated evidence base and is out of scope for this paper.

4.2 Experimental Pipeline

Every reported result is produced by the same nine-stage pipeline, run once per query under each combination of dataset, vote-construction regime, and evaluation cell studied:

1. obtain stored per-document scores from each base ranker (Section 4.4) for a fixed candidate pool;
2. normalize scores per query and per ranker (Equation (1));
3. derive weighted pairwise votes from normalized score margins (Section 3.2);
4. construct the query-specific weighted directed preference graph G_q under one vote-construction regime (Section 4.5);
5. extract unrepaired rankings directly from G_q and from the prior score alone;

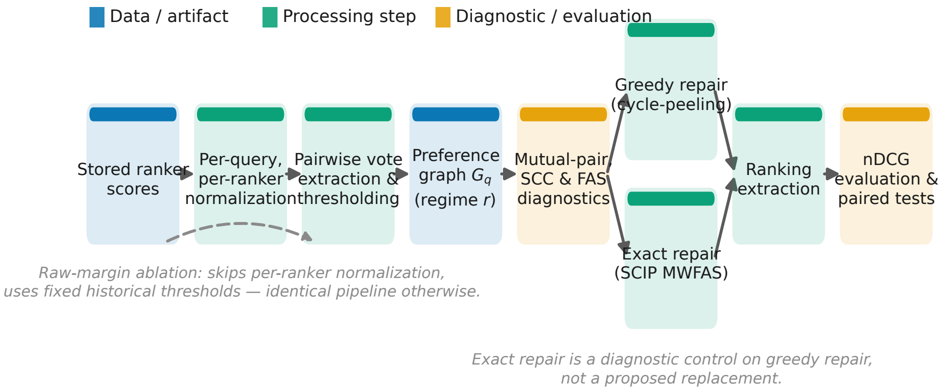

Fig. 1 Pipeline schematic. Stored ranker scores are normalized, converted into pairwise votes, and aggregated into a weighted preference graph, which is diagnosed for cycles and then either left unrepaired or repaired by greedy cycle-peeling or exact SCIP-based MWFAS solving – two co-equal branches, not a primary method and a footnote. Every branch feeds the same ranking-extraction and evaluation stages, so repaired-versus-unrepaired and greedy-versus-exact comparisons hold every downstream choice fixed.

6. repair G_q into $\widetilde{G}_q$ with greedy cycle-peeling or exact SCIP-based MWFAS solving (Section 4.6);
7. extract repaired rankings from $\widetilde{G}_q$ using the same extraction rule applied to G_q (Section 4.7);
8. evaluate every extracted ranking against relevance judgments with nDCG at a prespecified cutoff (Section 4.9); and
9. conduct paired, multiplicity-corrected statistical inference on repaired-versus-unrepaired differences (Section 4.10).

Stages 1–4 are shared by every downstream comparison; stages 5–9 are repeated once per extraction rule and once per repair method (unrepaired, greedy, exact) being compared. No stage regenerates upstream retrieval: stage 1 reuses fixed, versioned score files, so every reported statistic is reproducible from stored intermediates without repeating retrieval or querying a paid API (Section 4.11).

Figure 1 summarizes this pipeline, with the greedy and exact repair branches drawn explicitly as parallel paths through the same downstream extraction and evaluation stages, so that any difference between repaired and unrepaired rankings, or between greedy- and exact-repaired rankings, is attributable to the repair step alone.

4.3 Datasets and Retrieval Tasks

The study uses four public retrieval benchmarks, chosen to span distinct retrieval domains so that construction-sensitivity and repair findings can be checked for domain generality rather than being tied to one corpus type: SciDocs, a scientific-document

Table 1 Datasets and prespecified evaluation settings. Stored: usable query count after relevance-judgment filtering. Depth: verified common complete stored score depth across all three base rankers. Canonical cells use pool size P equal to the metric cutoff k; larger-pool cells use $P > k$, so that repair has room to change top-k set membership (Section 3.4).

Dataset	Usable	Depth	Evaluated (P, k) cells
SciDocs	120	50	canonical $(20, 20)$; larger-pool $(20, 5)$, $(20, 10)$, $(20, 20)$, $(50, 5)$, $(50, 10)$, $(50, 20)$
FiQA	120	50	same as SciDocs
HotpotQA	52	35	canonical $(10, 10)$; larger-pool $(10, 5)$, $(35, 5)$, $(35, 10)$, $(35, 20)$
BRIGHT	50	50	same as SciDocs

retrieval benchmark built around citation-informed document representations [7]; FiQA, a financial-opinion question-answering and retrieval benchmark [31]; HotpotQA, a multi-hop question-answering benchmark used here as a supporting-fact retrieval task [54]; and BRIGHT, a reasoning-intensive retrieval benchmark whose queries require inference beyond lexical or semantic similarity to identify relevant documents [44], accessed via its public `examples` configuration. SciDocs, FiQA, and HotpotQA also appear in general zero-shot retrieval benchmarking [46]; BRIGHT is included specifically for a more recent, reasoning-oriented retrieval setting not covered by that benchmark suite.

For every dataset, base-ranker scores, query-ID lists, and relevance judgments are reused unchanged from fixed, versioned stored files; no upstream retrieval is regenerated for this study. Queries are filtered only for relevance-judgment eligibility (a query must have at least one qrels-eligible relevant document among candidates that at least one ranker scores); this excludes 18 of HotpotQA's 70 stored queries and leaves every other dataset's stored queries usable. Candidate pools are formed, per query, from the union of documents scored by at least one base ranker, truncated to a fixed depth by the primary candidate-pool policy (Section 4.5); Table 1 reports the verified common stored score depth per dataset and the evaluation cells this study evaluates.

4.4 Base Rankers and Judgment Sources

Three base rankers provide heterogeneous ranking signals for the preference-aggregation study; they are a controlled substrate for studying construction and repair, not a claim of retrieval state of the art. **BM25** scores each candidate with the standard Okapi BM25 term-weighting formula [40], using the conventional parameter values $k_1 = 1.5$, $b = 0.75$. **TF-IDF** scores candidates by cosine similarity between query and document term-frequency–inverse-document-frequency vectors under standard logarithmic term-frequency weighting [41]. **MiniLM** scores candidates by cosine similarity between query and document embeddings from the pretrained `all-MiniLM-L6-v2` sentence-embedding model, a distilled transformer encoder [53] used here as a fixed, off-the-shelf dense-similarity signal, not fine-tuned for any dataset in this study. All three rankers are applied to the same per-query candidate pool and produce one real-valued score per candidate; higher scores indicate stronger estimated relevance for every ranker, so no ranker requires a direction transformation before normalization (Equation (1)).

4.5 Preference-Graph Construction

The primary construction protocol normalizes every ranker's scores with per-query, per-ranker min-max scaling (Equation (1)) and extracts directional votes from normalized score margins, following Section 3.2. Vote retention within a vote-construction regime uses a matched threshold: the per-ranker vote-margin threshold and the regime-level aggregate weight threshold are each chosen so that the resulting retained-vote rate matches the rate obtained from an unnormalized (raw-margin) construction on the same graphs, so that comparisons between the normalized primary protocol and the raw-margin ablation (Section 5) are not confounded by a change in how permissive thresholding is. Support requirements are fixed per regime: `ms2` requires support from at least two of the three rankers and a positive aggregate margin; `ms1` and `ms1_drop_mutual` require support from at least one ranker, differing only in whether mutual (two-way) contradictions are removed before aggregation (Section 3.2).

Candidate pools use one primary policy: for each query, the top-P documents of a reciprocal-rank-fusion ranking computed over the union of the three rankers' native (unnormalized) score lists. This "RRF-centered" pool policy is one specific, named choice among several equally valid ways to truncate a candidate pool, and it determines which documents are eligible to appear in D_q at all – a decision made once, upstream of score normalization and graph construction. Four alternative, independently defined pool policies (equal per-ranker depth, round-robin interleaving across rankers, a single-ranker pool, and a CombSUM-ranked pool) are evaluated as a robustness check (Section 5) to test whether the study's structural and retrieval conclusions depend on this specific choice.

Algorithm 1 summarizes graph construction for a single query, reflecting the procedure implemented in the repository's graph- and vote-construction code. The description is intentionally compact: the contribution is not a new preference extractor, but a controlled study of how these construction choices affect cyclicity, repair, and downstream retrieval.

4.6 Repair Methods

Two repair methods solve Equation (3), reported as co-equal throughout, per Section 3.4. **Greedy repair** is the repository's own deterministic cycle-peeling heuristic: while the graph contains a directed cycle, one cycle is located and its minimum-weight edge is removed, repeating until the graph is acyclic. This is not the classical Eades–Lin–Smyth linear-arrangement heuristic [11] or another named heuristic from the feedback-arc-set literature (Section 2.3); it is a simpler, repository-specific procedure, stated here as such rather than attributed to prior work it does not implement. **Exact repair** solves the linear-ordering mixed-integer program (Equation (4)) with the open-source SCIP solver via PySCIPOpt (version 6.2.1), requiring a proven-optimal status (relative optimality gap 0.0) rather than accepting

Algorithm 1 Preference-graph construction for one query

Require: Query q; candidate set D_q; base rankers $S = \{\text{BM25}, \text{TF-IDF}, \text{MiniLM}\}$ with scores $x_{q,s}$; construction regime $r \in \{\texttt{ms2}, \texttt{ms1}, \texttt{ms1\_drop\_mutual}\}$
Ensure: Weighted directed preference graph $G_q = (V_q, E_q, w_q)$

$V_q \leftarrow D_q$; initialize an empty weighted edge accumulator
for each ranker $s \in S$ **do**
  Normalize $x_{q,s}$ to $\bar{x}_{q,s}$ (Equation (1))
  for each unordered candidate pair $\{u, v\} \subseteq D_q$ scored by s **do**
    if $|\bar{x}_{q,s}(u) - \bar{x}_{q,s}(v)|$ is below the vote-margin threshold, or s does not score both u and v **then**
      Abstain on $\{u, v\}$ for ranker s
    else
      Accumulate directional support for $u \succ v$ or $v \succ u$ (whichever $\bar{x}_{q,s}$ supports), weighted by the normalized margin
    end if
  end for
end for
Apply regime r's support rule and aggregate weight threshold to every accumulated directed pair
if $r = \texttt{ms1\_drop\_mutual}$ **then**
  Remove every unordered pair retained in both directions
end if
$E_q \leftarrow$ every retained directed pair with positive aggregated weight
return $G_q = (V_q, E_q, w_q)$

a feasible incumbent, with a 300-second per-query wall-clock time limit as a safety margin. Every exact-repair result reported in this study reached proven optimality; the time limit was never approached (observed per-query solve times range from sub-millisecond to a few hundred milliseconds at the graph sizes studied here, several orders of magnitude below the limit). The exact solver's linear-ordering formulation was independently cross-checked against an unrelated brute-force exact method (exhaustive permutation search) on synthetic and real query graphs, with all checked cases matching exactly, before being used to produce the reported evidence.

Exact repair is used here as a diagnostic control on greedy repair, not as a proposed practical replacement for it: it is not evaluated for scalability beyond the query-graph sizes studied here, and this paper makes no general claim about exact MWFAS solving at larger scale. Its role is to answer RQ4's second half directly: because the exact solver's removed feedback arc set is, by construction, a global optimum of Equation (3), no repair method can remove less weight while still restoring acyclicity, so if exact repair's downstream retrieval effect is statistically indistinguishable from greedy repair's, greedy suboptimality is ruled out as an explanation for any observed retrieval null (Section 5).

4.7 Ranking Extraction Methods

Extraction methods fall into four groups, extending Section 3.5's general definitions with the specific methods this study evaluates and stating precisely which methods enter which comparison family (Section 4.8).

Prior-only and graph-free fusion rank candidates without constructing or consulting a preference graph: the prior-only method ranks by the normalized, summed base-ranker score used in Equation (5); reciprocal rank fusion, CombSUM, and Borda-style fusion combine the three rankers' individual per-query rankings or scores directly [8, 16].

Graph-only extraction computes a score from G_q or $\widetilde{G}_q$ alone. The Copeland score s_{cop} and balance score s_{bal} (Section 3.5) are evaluated on every dataset, regime, and repair method. Two related but distinct Markov-chain scores are also evaluated: a damped stationary-distribution score computed with a PageRank-style restart term, referred to throughout as *Markov*; and an undamped stationary distribution of a Markov chain built so that ergodicity is guaranteed by explicit self-loops rather than restart mass, following Rank Centrality exactly [33] and referred to as *Rank Centrality*. Standard graph PageRank (computed on the reversed graph, so that being preferred by highly-ranked competitors raises a node's score) [36] is evaluated as a further graph-only rule. A Bradley–Terry score is evaluated as a probabilistic alternative to these combinatorial and spectral rules: strengths $p_u > 0$ are fit by maximum likelihood under

$$P(u \succ v) = \frac{p_u}{p_u + p_v}, \tag{7}$$

using the iterative minorization-maximization (MM) algorithm, with the retained graph's edge weights as (possibly fractional) win counts [4, 20]. Finally, two ordering-based rules apply only to repaired (necessarily acyclic) graphs, since they are undefined on a cyclic input: a topological ordering of $\widetilde{G}_q$, and a priority-topological ordering that breaks ties among currently-available source nodes using the prior score.

Hybrid prior-plus-graph extraction combines a graph-only score with the normalized prior score via Equation (5). Copeland and balance hybrids are evaluated at a fixed mixing weight $\alpha = 0.3$ in the primary analysis, with a sweep over $\alpha \in \{0.1, 0.3, 0.5, 1.0\}$ reported as a robustness check; a Markov hybrid is evaluated at the same fixed α as an additional method in the expanded family (Section 4.8).

Every graph-dependent rule (graph-only and hybrid) is computed on both the unrepaired graph G_q and the repaired graph $\widetilde{G}_q$ wherever both are defined, holding the extraction rule fixed so that a repaired-versus-unrepaired difference is attributable to repair and not to a change of scoring rule. This directly addresses a prior reviewer concern that a fixed hybrid-fusion weight could suppress a genuine graph-repair effect: because graph-only rules ($\alpha = 1$ in spirit, no prior component) are evaluated alongside the hybrid rules, a repair effect masked by prior-score mixing at $\alpha = 0.3$ would still be visible in the graph-only comparisons and in the α-sweep.

4.8 Baselines and Comparisons

Table 2 summarizes every method's role. Two overlapping but distinct method families are evaluated, and Section 5 reports which family backs which specific comparison

Table 2 Method families evaluated in this study. Family: which comparison(s) a method backs (Section 5); primary = greedy-repair canonical/larger-pool comparisons; expanded = exact-repair canonical/larger-pool comparisons; struct. = exact-vs-greedy structural comparison only. Repair: which repair variants are compared for that method (U = unrepaired, G = greedy, E = exact).

Method	Family	Graph?	Prior?	Repair variants compared
Prior-only	primary, expanded	no	yes	n/a (no graph)
RRF	primary, expanded	no	no	n/a (no graph)
CombSUM	primary, expanded	no	no	n/a (no graph)
Borda fusion	primary, expanded	no	no	n/a (no graph)
Copeland	primary	yes	no	U, G, E
Balance	primary	yes	no	U, G, E
Markov (damped)	primary	yes	no	U, G, E
Copeland hybrid	primary	yes	yes (α=0.3)	U, G, E
Balance hybrid	primary	yes	yes (α=0.3)	U, G, E
PageRank	expanded	yes	no	U, E
Rank Centrality	expanded	yes	no	U, E
Markov hybrid	expanded	yes	yes (α=0.3)	U, E
Bradley–Terry	expanded	yes	no	U, E
Topological	struct. only	yes	no	G vs. E only
Priority-topological	struct. only	yes	yes (tie-break)	G vs. E only

rather than pooling them: a *primary family* (four graph-free baselines; Copeland, balance, and Markov as graph-only and hybrid rules) backs the primary canonical and larger-pool greedy-repair-versus-unrepaired comparisons; an *expanded family* adds PageRank, Rank Centrality, a Markov hybrid, and Bradley–Terry specifically to the exact-repair-versus-unrepaired canonical and larger-pool comparisons, so that the exact-repair diagnostic control (Section 4.6) is not limited to the extraction rules the primary family happens to use. Topological and priority-topological ordering are evaluated only in a separate exact-versus-greedy structural comparison, since they require an already acyclic graph and so have no unrepaired counterpart.

A "strong baseline" here means the strongest method actually evaluated in this study, not a method added rhetorically: CombSUM and reciprocal rank fusion are simple, graph-free, and require no repair step at all, so their inclusion tests whether any graph-based method (repaired or not) earns its added complexity relative to methods that discard graph structure entirely. None of the methods in Table 2 is presented as state-of-the-art retrieval; they are a controlled comparison set for isolating the effect of graph construction and repair, not a search for the best possible ranker.

4.9 Evaluation Metrics

The primary retrieval metric is normalized discounted cumulative gain (nDCG) at a prespecified cutoff k (Equation (6)), evaluated per query and averaged across a dataset's usable queries; higher values indicate better agreement with graded relevance judgments. Mean reciprocal rank (MRR) and mean average precision (MAP) are computed as secondary diagnostics only and are never used as a primary confirmatory metric. All three are computed at the same evaluated (P, k) cell (Table 1) for every method and repair variant being compared.

Three kinds of quantity are kept explicitly distinct throughout this study, restating Section 3.5's closing point in operational terms: the *repair objective* (removed feedback-arc weight, Equation (3)) is a graph-internal optimization target; *structural diagnostics* (cyclic-query rate, strongly connected component structure, removed weight as a fraction of total graph weight, and top-k membership-change rate between repaired and unrepaired rankings) describe what repair changed in the graph; and *retrieval metrics* (nDCG, MRR, MAP) describe agreement with external relevance judgments. No structural diagnostic is treated as a proxy for, or averaged together with, a retrieval metric at any point in this study's analysis.

4.10 Statistical Analysis

The unit of statistical inference is the query: for a given dataset, construction regime, evaluation cell, and extraction method, the repaired-minus-unrepaired nDCG difference is computed once per query, and all inference is conducted on this vector of paired per-query differences, never on a coarser or finer unit. The primary confirmatory test for every repaired-versus-unrepaired comparison is an exact sign-flip permutation test on the paired differences (enumerating all 2^m sign patterns when the number of nonzero differences $m \leq 20$; a 10,000-replicate Monte Carlo approximation with a fixed seed otherwise), testing the null hypothesis that the paired differences are symmetric about zero. Every p-value from a prespecified family of cells (e.g. the full canonical family, or the full larger-pool family) is adjusted jointly with the Holm step-down procedure [18], controlling the family-wise error rate at $\alpha = 0.05$; different families (canonical, larger-pool, exact-repair, alternative-pool, added-baseline) are never pooled into one joint correction unless stated otherwise, so that a family's size reflects only the comparisons it was prespecified to contain.

Beyond this primary test, four robustness procedures are reported for every primary family, never as a substitute for the Holm-corrected decision: percentile, basic, and bias-corrected-and-accelerated (BCa) bootstrap 95% confidence intervals on the mean paired difference [12]; leave-one-out influence analysis, removing one query at a time to check whether any single query drives a family's conclusion; the Benjamini–Hochberg false-discovery-rate procedure [3] as an alternative, less conservative correction, reported alongside (not in place of) the primary Holm decision; and a minimum-detectable-effect (MDE) analysis at 80% power, and two-one-sided-tests (TOST) narrow-equivalence testing at prespecified margins, both used only to characterize what the study's sample size can and cannot rule out, not to claim equivalence has been established (Section 2.4).

For the appendix-scoped real-large-language-model pilot (Section 1), the unit of inference is the query *cluster*: six queries, each replicated across several provider, paradigm, and repetition conditions, are treated as six independent clusters, never as row-level independent observations. Row-level analysis of this pilot's replicated rows (correct unit: $n = 6$; incorrect unit: one row per replicate) systematically overstates the pilot's evidential weight and is not used anywhere in this study.

4.11 Reproducibility and Implementation

The full pipeline (Section 4.2) is implemented in Python, using `networkx` for graph representation and cycle detection, `numpy`/`scipy` for numerical and statistical routines, and `PySCIPOpt` 6.2.1 (open-source SCIP) for exact repair (Section 4.6); the MiniLM ranker (Section 4.4) uses the `sentence-transformers` library to load the fixed, pretrained `all-MiniLM-L6-v2` checkpoint. Every Monte Carlo or bootstrap procedure (Section 4.10) uses a fixed random seed, so that reported p-values and intervals are exactly reproducible from the same stored inputs. No stage of the pipeline used to produce a main-paper statistic requires a paid API call: base rankers use fixed, versioned, locally computed or stored score files, and the appendix-scoped real-LLM pilot (which does call paid provider APIs) is never a source of a main-paper number. All code, stored intermediates, and the scripts used to regenerate every reported statistic and table are available at the repository referenced in Section 9.

Generative-AI assistance. Generative artificial-intelligence tools (Anthropic's Claude, via the Claude Code command-line environment) were used to assist with software development for this study – writing and revising analysis code, statistical-inference utilities, figure-generation scripts, and audit scripts – and with drafting, revising, and fact-checking portions of this manuscript's text and figures. Every AI-assisted code change, statistical analysis, figure, table, citation, and passage of manuscript text was reviewed and independently verified against the underlying stored data and source code by the author before inclusion; no generative-AI output was accepted without that verification. The author takes full responsibility for the accuracy, integrity, and originality of the source code, experiments, statistical analyses, figures, tables, interpretations, citations, and manuscript text in this work. No generative-AI tool is credited as an author, and no generative-AI tool independently designed the study, selected the research questions, or made scientific judgments attributed to the author in this manuscript.

5 Results

This section reports what the canonical evidence package shows for each research question in turn (Section 4.1): how much structural inconsistency the derived preference graphs contain (RQ1), what repair changes structurally (RQ2), whether that change is accompanied by a statistically supported retrieval-effectiveness gain (RQ3), and whether the answer to RQ3 depends on repair-method optimality or on the construction, pooling, and extraction choices held fixed elsewhere in the study (RQ4). Every statistic below is computed on the canonical `primary_minmax_retention_matched` protocol (Section 4.5) unless stated otherwise, and every significance statement uses the Holm-corrected decision at $\alpha = 0.05$ from the prespecified family it belongs to (Section 4.10); raw, uncorrected p-values are reported only alongside the corrected decision, never as a substitute for it.

Table 3 Structural inconsistency by dataset and vote-construction regime, canonical protocol. Cyclic: fraction of queries whose graph contains a directed cycle. Post-mutual: the same fraction after direct two-way contradictions are removed before aggregation (Section 3.2). Norm. FAS wt.: mean feedback-arc weight removed by repair, as a fraction of total graph weight.

Dataset	Regime	Cyclic	Post-mutual	Norm. FAS wt.
SciDocs	`ms2`	0.0%	0.0%	0.000
	`ms1`	99.2%	10.8%	0.074
	`ms1_drop_mutual`	11.7%	11.7%	0.002
FiQA	`ms2`	0.0%	0.0%	0.000
	`ms1`	98.3%	30.8%	0.080
	`ms1_drop_mutual`	30.8%	30.8%	0.004
HotpotQA	`ms2`	0.0%	0.0%	0.000
	`ms1`	63.5%	1.9%	0.029
	`ms1_drop_mutual`	1.9%	1.9%	0.000
BRIGHT	`ms2`	0.0%	0.0%	0.000
	`ms1`	92.0%	22.0%	0.055
	`ms1_drop_mutual`	22.0%	22.0%	0.003

5.1 Structural Inconsistency Before Repair

The vote-construction regime, not the underlying documents, determines whether a query's preference graph is cyclic at all. Table 3 reports, for every dataset and regime, the fraction of queries whose graph contains a directed cycle, that same fraction after direct mutual contradictions are removed, and the mean feedback-arc weight repair would need to remove, expressed as a fraction of total graph weight.

The conservative regime, `ms2`, is acyclic on every query in every dataset in the observed evidence (Section 3.2): repair has no graph to act on under this regime. The permissive regime, `ms1`, is cyclic on the large majority of queries in every dataset, ranging from 63.5% (HotpotQA) to 99.2% (SciDocs). Most of that cyclicity is direct, two-way contradiction rather than longer-range nontransitivity: removing mutual pairs before aggregation (`ms1_drop_mutual`) reduces the cyclic-query rate to between 1.9% (HotpotQA) and 30.8% (FiQA), and `ms1_drop_mutual`'s own cyclic rate matches `ms1`'s post-mutual rate exactly in every dataset, confirming that the residual cyclicity in both cases is the same underlying longer-cycle structure. Figure 2 shows this decomposition across datasets.

Raw, unnormalized score margins are not a scale-neutral starting point for this construction (Section 3.1). Under the raw-margin ablation, BM25 contributes a mean conditional edge-weight share of 0.988 whenever it participates in an edge; under the primary normalized protocol, that share falls to 0.512. Figure 3 shows this raw-versus-normalized contrast holds across all four datasets and all three vote-construction regimes: a graph built from raw margins is close to a single-ranker graph with multi-ranker labels attached, and normalization is what restores a genuinely mixed graph.

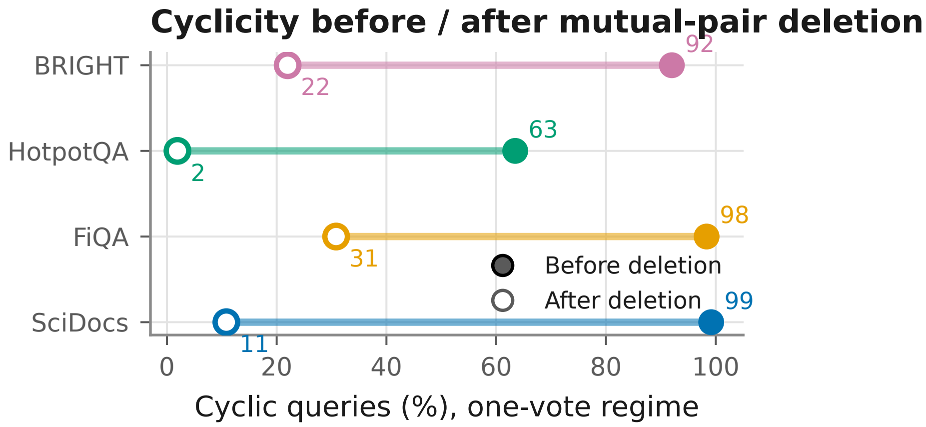

Fig. 2 Cyclic-query percentage under `ms1` before and after deleting direct mutual pairs, by dataset. Most active-regime cyclicity is direct two-way contradiction, not longer-cycle structure, but a nonzero, dataset-dependent residual remains after mutual-pair deletion.

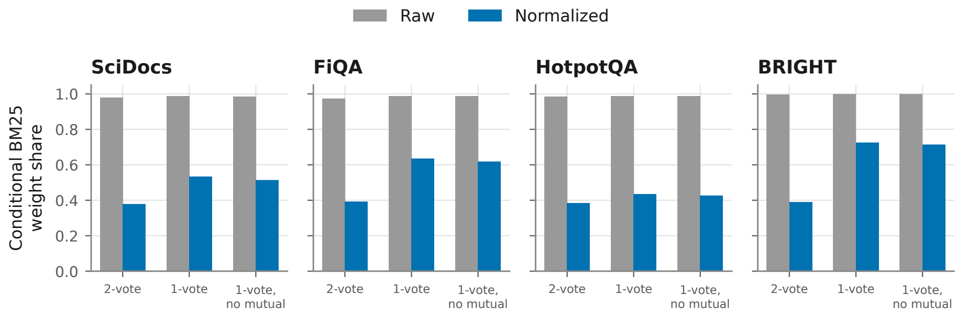

Fig. 3 Conditional BM25 edge-weight share under the raw-margin ablation versus the primary normalized protocol, by dataset and vote regime. Raw construction is near-total BM25 domination in every panel; normalization restores a mixed graph.

5.2 Structural Effect of Repair

Repair is structurally active precisely where Section 5.1 shows a graph has contradictory structure to remove, and inert elsewhere. Under `ms2`, repair removes nothing on any query, because no graph is cyclic. Under `ms1`, the active regime, repair removes a mean of 7.4% (SciDocs) to 8.0% (FiQA) of total graph weight; under `ms1_drop_mutual`, the fraction still to remove after mutual pairs are already gone drops to 0.2%–0.4% (Table 3). This confirms that graph construction, not the repair algorithm, is what decides whether repair has any opportunity to act at all – a point

Table 4 Retrieval effectiveness under greedy repair, canonical protocol. Top: Holm-corrected repaired-vs-unrepaired nDCG comparison families (5-pair primary method family, Section 4.8). Bottom: dataset-macro mean nDCG for selected methods, for context (not a significance claim; graph-free baselines require no repair step at all).

Comparison family	Cells	Holm-rejected	Smallest Holm-adjusted p
Active canonical (`ms1`)	20	0/20	0.384
Full canonical (all regimes)	60	0/60	0.384
Active larger-pool (`ms1`, $P>k$ incl.)	110	0/110	0.352

Method (dataset-macro mean nDCG)	Mean nDCG
CombSUM (graph-free fusion)	0.554
Reciprocal rank fusion (graph-free)	0.546
Copeland hybrid, repaired (`ms1`)	0.546
Prior-only (graph-free)	0.545
Borda fusion (graph-free)	0.545

kept separate here from whether that activity translates into a retrieval effect, which Section 5.3 addresses directly.

Repair also changes which documents are visible for evaluation once the candidate pool exceeds the metric cutoff. At $P = k$, top-k set membership cannot change by construction: the evaluated prefix is the entire pool, so repairing the graph can reorder documents but never add or remove one from the top-k set, and the measured membership-change rate is exactly 0%. Once $P > k$, pooling across all three vote-construction regimes and every larger-pool evaluation cell (Section 4.3), the mean top-k membership-change rate rises to 10.6%. Repair is therefore not merely reordering an already-fixed prefix under $P > k$: it can and does change which documents are evaluated at all. Whether that structural activity is accompanied by a reliable retrieval-effectiveness gain is addressed next.

5.3 Retrieval Effectiveness

Table 4 reports the Holm-corrected outcome for every prespecified repaired-versus-unrepaired nDCG comparison family under greedy repair, together with dataset-macro mean nDCG for the strongest graph-free baselines and the best-performing repaired hybrid, for context.

No repaired-versus-unrepaired nDCG comparison survives Holm correction in any of the three prespecified families: not the 20-cell family restricted to the structurally active `ms1` regime, not the full 60-cell canonical family pooling all three regimes, and not the 110-cell active larger-pool family that includes every $P > k$ cell where repair has room to change top-k membership (Section 5.2). The smallest Holm-adjusted p-value in the canonical families is 0.384 (SciDocs, Copeland graph, raw sign-flip $p = 0.020$ before correction); in the larger-pool family it is 0.352. Figure 4 shows the corresponding paired bootstrap intervals for the active `ms1` regime: point estimates are sometimes positive, but the great majority of intervals cross zero, consistent with the Holm decision.

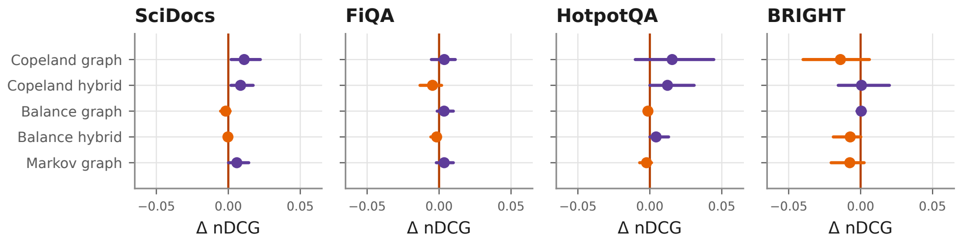

Fig. 4 Repaired-minus-unrepaired mean ΔnDCG with paired 95% bootstrap intervals, active `ms1` regime, primary five-pair method family. Point estimates vary in sign across datasets and methods, and nearly every interval includes zero.

Simple graph-free baselines remain competitive with the best repaired graph-dependent method throughout: CombSUM's dataset-macro mean nDCG (0.554) exceeds every graph-dependent method's, and reciprocal rank fusion (0.546) is effectively tied with the best-performing repaired hybrid, the `ms1` repaired Copeland hybrid (0.546). Repair is therefore not shown to produce a retrieval ranking that outperforms simple score or rank fusion under the canonical protocol.

5.4 Exact Versus Heuristic Repair

The comparisons in Section 5.3 use greedy repair, which does not generally reach the optimum of the feedback-arc-set objective (Equation (3)). This subsection reports what changes when repair is instead solved to certified optimality, addressing RQ4's second half directly: whether the retrieval null in Section 5.3 could instead reflect greedy repair's suboptimality.

Every exact solve in the canonical evidence package reached proven optimality: 1,025/1,025 query graphs, relative optimality gap 0.0, with a mean solve time of 7.4ms (median 0.25ms, maximum 236ms) against a 300-second per-query safety limit that was never approached (Section 4.6). Restricted to the 379 queries (pooled across all three vote-construction regimes) whose raw graph is actually cyclic, exact repair finds a different feedback arc set than greedy in 87.9% of them, and removes systematically less total edge weight: Table 5 and Figure 5 report the per-dataset gap. Greedy cycle-peeling is therefore a real, non-trivial overestimate of the minimum weighted feedback arc set on this evidence, not merely a theoretical possibility.

That structural gap does not translate into a retrieval-level difference. Pooled across all seven repaired-graph-dependent extraction rules evaluated in this comparison (Copeland, balance, Markov, both hybrids, topological, and priority-topological orderings; Section 4.7) and all five validated metrics, none of the 35 pooled metric-method cells is Holm-significant; at the finer 399-cell per-dataset-and-regime granularity, none is Holm-significant either. The largest raw, uncorrected effect in the pooled family is a mean nDCG@10 difference of -0.00223 for priority-topological ordering (raw sign-flip $p = 0.026$; Holm-adjusted $p = 0.906$).

Table 5 Exact versus greedy repair, structural comparison, cyclic queries only ($n = 379$, pooled across vote-construction regimes). Mean weight removed is reported for both repair methods; the last column is the relative reduction from greedy to exact. The corresponding retrieval-level comparison (whether exact- and greedy-repaired rankings differ in nDCG, pooled across every repaired-graph-dependent extraction rule and validated metric – nDCG@5/10/20, MRR, MAP, 1,025 queries per cell, all regimes) finds 0/35 pooled and 0/399 per-dataset-and-regime cells Holm-significant; no dataset breakdown applies to that pooled result, so it is not repeated per row here.

Dataset	Wt. removed (exact)	Wt. removed (greedy)	Reduction
SciDocs	6.14	8.19	25.1%
FiQA	4.51	6.37	29.2%
HotpotQA	1.55	1.80	13.4%
BRIGHT	3.07	3.95	22.3%

Exact repair also does not change the headline comparison against the unrepaired graph. Using the expanded nine-method family that adds PageRank, Rank Centrality, a Markov hybrid, and Bradley–Terry to the primary five-pair family specifically for this comparison (Table 2), 0 of 36 canonical exact-repaired-versus-unrepaired cells and 0 of 56 larger-pool cells are Holm-significant – the same qualitative outcome as the greedy-versus-unrepaired comparison in Section 5.3, now confirmed under a repair method that cannot be outperformed on the repair objective itself by any other feasible solution. Because exact repair improves the graph objective (Equation (3)) more than greedy repair does, yet does not reveal a retrieval gain greedy repair missed, heuristic suboptimality is ruled out as the explanation for the retrieval null in Section 5.3.

5.5 Robustness

Table 6 collects four canonical robustness checks against the primary retrieval finding, each drawn from a prespecified comparison family rather than a post-hoc search.

Each check addresses a distinct way the primary null result could have been an artifact of one specific design choice – an advantageous candidate pool, insufficient statistical power, or a $P = k$ evaluation that structurally hides repair's effect – and none overturns it. The power analysis is the one qualification worth stating precisely: this evidence rules out a general retrieval effect of the size the point estimates themselves suggest (median absolute delta 0.0036), but a considerably smaller general effect, below the 0.02-nDCG detectable scale, is not ruled out by this sample size.

5.6 Supporting LLM Evidence

A bounded pilot supplements the score-derived evidence above with real large-language-model pairwise judgments: six queries, four providers, and four preference-graph construction variants, analyzed at the query-cluster level ($n = 6$ independent clusters), since the pilot's replicated rows within a query are repeated measurements, not additional independent samples. Whole-graph repair's effect on retrieval, estimated the same way as the score-derived comparisons above, is a mean nDCG change

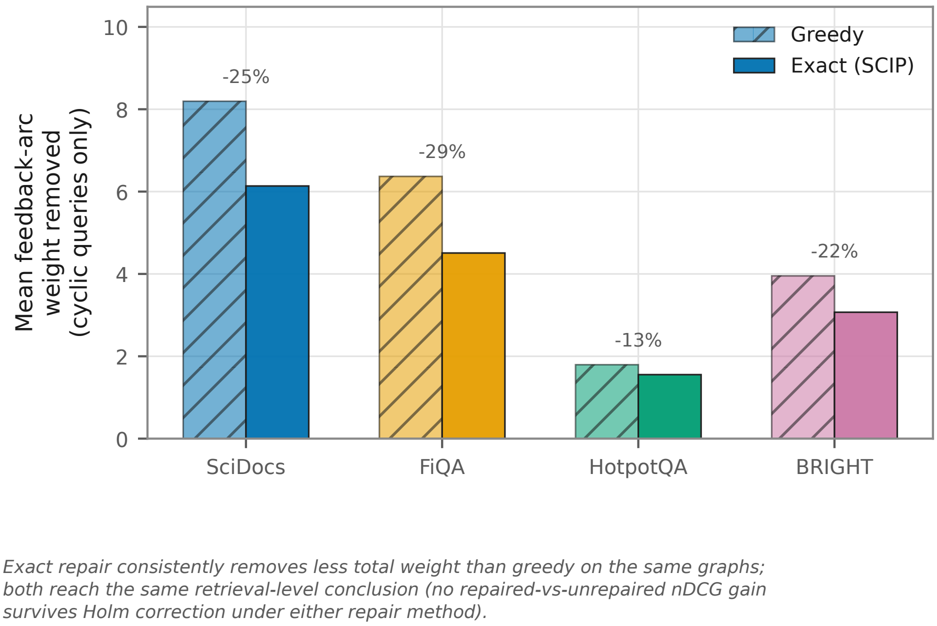

Fig. 5 Mean feedback-arc weight removed by greedy versus exact repair, cyclic queries only, by dataset. Exact repair removes less weight than greedy in every dataset, confirming that greedy over-removes rather than under-removes edge weight relative to the true optimum.

of 0.00019 with a cluster-level 95% interval of $[-0.00072, 0.00140]$ (exact sign-flip $p = 0.875$): indistinguishable from zero, directionally consistent with the null finding in Section 5.3. Six clusters give this pilot very limited power, so this result is reported as directional and supporting only – it does not, on its own, extend any conclusion in this paper beyond the four benchmarks and construction regimes making up the main evidence package (Section 4.3).

6 Discussion

6.1 Main Findings

Four findings, taken together, answer the research questions stated in Section 4.1. Preference-graph repair causes genuine structural change: under the permissive vote-construction regime, both greedy and exact repair remove a measurable share of contradictory edge weight, and once the candidate pool exceeds the evaluation cutoff, repair changes which documents even enter the evaluated top-k set at a non-negligible rate (Section 5.1, 5.2). Exact optimization confirms and sharpens that structural finding rather than overturning it: solved to certified optimality, repair removes systematically less edge weight than the greedy heuristic on the same graphs, showing that greedy over-corrects relative to the true minimum-weight feedback arc set

Table 6 Robustness checks against the null retrieval finding (Section 5.3), canonical evidence only.

Check	Result
Baseline fairness	Testing whether the RRF-centered primary candidate pool gives a structural advantage to RRF-family baselines: 0/8 Holm-significant cells (RRF and CombSUM vs. the fixed repaired Copeland hybrid, canonical pool, all four datasets). No evidence of a pool-driven advantage.
Power and minimum detectable effect	In the active larger-pool family, median observed absolute mean delta 0.0036 vs. median Holm-adjusted 80%-power minimum detectable effect 0.0201: the study is well positioned to rule out larger effects, but many smaller ones remain below the corrected detectable scale.
Narrow equivalence	Two-one-sided-tests equivalence, active larger-pool family: 13/110 cells reject non-equivalence at a ± 0.005 margin, 32/110 at ± 0.010, after Holm correction. Equivalence is established for a minority of cells, not the majority; the correct reading of the null result is absence of reliable positive evidence, not blanket practical equivalence.
Larger-pool ($P>k$) family	Already reported in Table 4: 0/110 Holm-significant despite a 10.6% mean top-k membership-change rate (Section 5.2), so membership change alone is not sufficient to produce a corrected retrieval gain.

(Section 5.4). Despite that genuine and now optimality-confirmed structural activity, no repaired-versus-unrepaired retrieval-effectiveness gain is statistically supported as a general effect under the canonical protocol: across three prespecified retrieval comparison families, four benchmarks, and both repair methods, none survives Holm correction (Section 5.3). Exact repair does not reverse that retrieval conclusion; the same null outcome holds whether the graph is repaired by a fast heuristic or by a solver that cannot be outperformed on the repair objective itself (Section 5.4).

Read together, these four findings support one conclusion: structural consistency and downstream retrieval utility behave as empirically distinct objectives in this evidence, not as the same underlying quantity observed at different resolutions. This is a narrower claim than it may first appear, and it is worth being precise about what it does and does not say. It does not say that graph repair never helps retrieval, that acyclicity is a useless property of a preference graph, that exact repair always fails to improve rankings, or that the null hypothesis of no effect has been proven; a null result under corrected inference is evidence against a general, reliably detectable effect at the scale this study is powered to observe, not a proof that no effect exists anywhere. The results are more accurately summarized as: repair did not yield a statistically supported general improvement in retrieval effectiveness under the canonical protocol, exact optimization does not change that conclusion, and small or dataset-specific effects below this study's detectable scale remain possible (Section 5.5's power analysis bounds how small). On the evidence gathered here, treating acyclicity as a surrogate for retrieval effectiveness is not supported.

6.2 Why Structural Improvement Need Not Improve Retrieval

The formulation in Section 3.5 already states the structural reason a repair-objective improvement need not track an nDCG improvement: the two objectives are computed from different information. The minimum-weight feedback-arc-set objective (Equation (3)) minimizes disagreement with the graph's own retained edges; nDCG scores agreement between an extracted ranking and external relevance judgments that never enter the repair objective. A design that makes one of these quantities smaller has no logical requirement to make the other larger. This subsection interprets that formal gap in light of what the results show, distinguishing what the experiments establish from what is a plausible mechanism consistent with, but not directly tested by, the experimental design.

Several mechanisms are consistent with a genuine structural repair producing no detectable retrieval effect, without any one of them being individually isolated by this study's design. Feedback-arc-set repair targets the least-weight edges needed to restore acyclicity (Section 3.4); by construction, those are the edges the graph itself treats as weakest evidence, which need not coincide with the edges separating a highly relevant document from a marginally relevant one near the top of a ranking. A cycle broken far from the top-ranked positions can leave the extracted ordering of the most relevant documents unchanged even though the graph's global structure is now acyclic. Where a hybrid extraction rule mixes a graph-derived score with a prior ranking signal (Equation (5)), a sufficiently informative prior can also compress the downstream influence of whatever the graph-side score does – consistent with, though not the sole explanation for, this study's observation that graph-only and hybrid methods do not diverge sharply in Section 5.3. Different extraction rules translate the same repaired graph into different rankings by different mechanisms (win-count differencing for Copeland, weighted margin differencing for balance, a stationary distribution for the Markov-chain rules; Section 4.7), so a structural change need not propagate identically through every rule, which is one plausible contributor to the mixed-sign point estimates in Figure 4. Finally, pairwise preference weights and relevance judgments are constructed from different evidence (Section 3.2, Section 3.1): a vote records one ranker's relative ordering of two candidates, not an external judgment of either candidate's relevance, so the graph's internal consistency and its agreement with relevance judgments are free to vary independently.

None of these four mechanisms is demonstrated as *the* cause of the null retrieval result; the experimental design compares repaired and unrepaired outcomes under corrected inference, not the causal pathway by which repair does or does not influence a given ranking. They are offered as mechanisms consistent with the design and the observed pattern of results, not as claims independently established by a dedicated test.

6.3 Exact Repair as a Diagnostic Control

A repair study that only ever applies a heuristic leaves one explanation for a null retrieval result permanently open: the heuristic may simply be too weak, so a better repair algorithm might have revealed the gain the weak one missed. This is not a hypothetical concern for cycle-peeling heuristics specifically – Section 5.4 shows exact repair removes systematically less edge weight than greedy on the same graphs, confirming that greedy is a real, non-trivial overestimate of the minimum feedback arc set here, not merely a theoretical possibility. That same section shows the retrieval-level conclusion is unchanged under exact repair: solved to certified optimality, a repair method that cannot be outperformed on the repair objective itself still does not reveal a retrieval gain greedy repair missed. Because optimality rules out suboptimality as an explanation by construction, this directly closes the heuristic-weakness alternative explanation for the retrieval null.

This role for exact SCIP-based repair is deliberately narrow. It is not introduced as a novel solver, a novel formulation of the feedback-arc-set problem, or a claim about how minimum-weight feedback-arc-set solving should be done at scale (Section 2.3 situates the formulation itself as long-studied, classical combinatorial optimization). Its contribution here is methodological control: closing a specific alternative explanation for a specific empirical finding, not proposing a better repair algorithm for practitioners to adopt.

6.4 Relation to Prior Literature

The finding that graph repair is structurally active but not reliably retrieval-improving sits against three adjacent, and non-contradictory, strands of prior evidence. First, pairwise comparison signals genuinely improve ranking quality in several settings: pairwise ranking prompting with large language models is competitive with, and in some settings better than, pointwise scoring for text reranking [39], and pointwise, pairwise, and listwise LLM reranking paradigms show real, measured effectiveness differences depending on formulation [45]. Those findings are about whether *using* pairwise evidence, instead of pointwise scores, changes ranking effectiveness – a different question from this paper's, which holds the pairwise evidence source fixed and asks whether *repairing* contradictions within that evidence changes effectiveness further. Nothing in this paper's results is in tension with pairwise comparison being an effective ranking signal; the null result is specifically about the marginal effect of consistency repair on top of an already-constructed pairwise-derived graph.

Second, prior work documents that pairwise judgments, including LLM-generated ones, are prone to inconsistency and order sensitivity: positional and prompt-order biases can affect LLM-as-judge outcomes [58], systematic position bias has been measured directly in LLM-as-judge pairwise comparisons [43], and intrinsic inconsistency in LLM-based pairwise rankings has motivated explicit aggregation methods to mitigate it [56]. This literature establishes that inconsistency is a real, measurable property of pairwise preference data, which is the premise this paper's construction-sensitivity results (Section 5.1) are consistent with. It does not by itself establish that *removing* that inconsistency improves a downstream task metric; that is the further, separate claim this paper tests directly.

Third, and closest to this paper's specific question, recent work on acyclic preference evaluation for language models treats acyclicity as a desirable property of an aggregated evaluation and reports downstream gains from enforcing it, on tasks including model ranking, response selection, and fine-tuning data selection [19]. That result is not contradicted by this paper: the two studies test acyclicity's downstream value in different task settings (LLM evaluation and data selection versus document retrieval) with different aggregation and repair mechanisms, so a positive effect in one setting and an unsupported general effect in another are both consistent with the same underlying fact – that whether structural consistency helps a downstream task is an empirical property of the specific task and pipeline, not a property of acyclicity itself that transfers automatically across settings. This paper's contribution relative to that literature is the controlled separation (Section 2.4): holding construction explicit, comparing unrepaired, heuristically repaired, and exactly repaired graphs, and testing the structural-to-utility link directly under corrected paired inference, for the retrieval setting specifically.

This paper is a substantial revision of an earlier preprint by the same author [48] (Section 2.3), which studied algorithmic ranking extraction from pairwise comparisons under a less conservative protocol and reported a conditional, dataset-dependent positive retrieval effect. The present study is retrieval-focused and controlled rather than algorithmic: it separates construction, repair, and extraction as independently varied factors, adds exact repair as a diagnostic control the earlier preprint did not include, and applies multiplicity-corrected paired inference throughout. The two should not be read as an earlier and later measurement of the same effect; the earlier preprint's specific numerical findings are superseded by this study's more conservative protocol, not confirmed by it.

6.5 Practical Implications

The evidence in this paper supports a small number of restrained, directly evidence-backed recommendations for researchers and practitioners building preference-graph pipelines.

- Do not assume an acyclic or less-cyclic preference graph is a better retrieval input without checking; this study's evidence does not support that assumption as a general rule.
- Evaluate repaired and unrepaired variants on the downstream task metric directly, with an unrepaired baseline always included as a comparison point, rather than reporting only structural diagnostics (cyclicity, removed weight) as if they were evidence of downstream benefit.
- Where a heuristic repair method is used and a null or weak downstream effect is observed, an exact solve on a representative subset of instances is an effective and specific check on whether heuristic suboptimality, rather than repair itself, could explain the result – this is a diagnostic use of exact solving, not a claim that exact repair must be run at full scale.

- Report graph-construction choices (normalization, vote semantics, candidate pooling), extraction rule, and evaluation cutoff explicitly alongside any repair claim: Section 5.1 shows these choices, not the repair algorithm, are what determine whether a graph has any cyclic structure for repair to act on in the first place.
- Treat structural-quality metrics and downstream-utility metrics as separate reporting dimensions with separate evidence, not as interchangeable indicators of pipeline quality.

These recommendations are audit practices, not a validated deployment rule for deciding when to apply repair. This paper does not establish, and does not attempt to establish, a criterion for predicting per-query whether repair will help; that is a distinct research question this paper's evidence base does not address.

6.6 Implications for LLM-Based Ranking

Pairwise judgments elicited directly from large language models are subject to the same structural possibility studied throughout this paper: aggregating several such judgments, or several repeated elicitations, can produce a preference graph with directed cycles, for the same reason score-derived votes can (Section 3.2). The conceptual distinction this paper's evidence supports – that structural consistency and downstream task utility are separate quantities requiring separate evidence – applies to that setting in principle. This paper's own LLM-derived evidence is a six-query bounded pilot (Section 5.6), directionally consistent with the main score-derived finding but far too small to establish a general claim on its own. The data gathered here do not support broad conclusions about modern LLM-based rerankers, pairwise-judgment pipelines at production scale, or any specific LLM reranking paradigm; extending this paper's controlled comparison to a larger, dedicated real-LLM evidence base is a direction for future work, not a claim this paper makes.

7 Limitations and Threats to Validity

7.1 Internal Validity

The pipeline (Section 4.2) makes several sequential design choices – score normalization, vote-margin and aggregate-weight thresholds, candidate pooling, and the hybrid mixing weight α – each of which could in principle interact with the repair effect being measured. Two controls mitigate this: the retention-matched thresholding procedure (Section 4.5) ties the normalized protocol's permissiveness to the raw-margin baseline rather than choosing it freely, and every retrieval comparison holds construction, pooling, and extraction fixed while varying only the repair method (Section 5.3), so a measured repaired-versus-unrepaired difference cannot be attributed to a confound in those other choices. What remains unresolved is α-dependence beyond the sweep reported as a robustness check (Section 4.7): the primary analysis fixes $\alpha = 0.3$, and while the sweep does not change the qualitative conclusion, it does not exhaustively rule out every possible mixing weight. The paired sign-flip and Holm-correction procedure (Section 4.10) assumes exchangeability of paired differences under the null and independence across queries within a family; the query-level pairing is designed to satisfy the first assumption, and queries are drawn from disjoint benchmark query sets to support the second, but formal independence across the three vote-construction regimes evaluated on the same underlying documents (Section 3.2) is not separately tested, since the canonical and larger-pool families are never jointly corrected across regimes for exactly this reason (Section 4.10).

7.2 Construct Validity

Acyclicity is one specific, narrow notion of structural consistency (Section 3.3); it is not the only property a preference graph could be judged by, and this study's structural diagnostics (cyclic-query rate, removed feedback-arc weight, top-k membership-change rate) are proxies for graph-internal coherence, not direct measurements of evidential correctness. nDCG is this study's primary utility construct; it captures rank-weighted agreement with graded relevance judgments, but it is one operationalization of retrieval quality among several (Section 4.9 reports MRR and MAP only as secondary diagnostics, precisely because the primary claim should not rest on whichever metric happens to be most favorable). A further construct gap, not resolved by this study's design, is that pairwise vote weight (Section 3.2) encodes a ranker's confidence in a relative ordering, not a judgment of either candidate's relevance; the two constructs are related in that both derive from the same underlying documents, but they are not the same measurement, and this study does not claim the graph's internal weights are a proxy for relevance at any point (Section 3.5's closing paragraph states this distinction as a matter of definition, not as an empirical finding requiring separate validation here).

7.3 External Validity

The main evidence base spans four benchmarks across four retrieval domains, two lexical base rankers, one dense base ranker, and three vote-construction regimes (Section 4.3, Section 4.4); this is a deliberately controlled, not exhaustive, sweep, chosen so that construction-sensitivity and repair findings could be checked for domain generality rather than tied to one corpus. It does not, by itself, establish that the findings generalize to retrieval domains, rankers, or construction regimes outside this set. The real-large-language-model evidence is a six-query pilot (Section 5.6), directionally consistent with but far too small to independently confirm the main finding, and this paper does not claim it does. This study does not evaluate learned fusion, cross-encoder reranking, modern dense rerankers beyond the single MiniLM checkpoint used here, or online (interactive, feedback-driven) retrieval evaluation; whether structural repair behaves differently under any of those settings is not addressed by this evidence and is not implied one way or the other. Extension to other preference-aggregation domains that share the same underlying structure – recommendation, model evaluation via pairwise LLM judging, or crowdsourced preference collection – is plausible in principle (Section 6.6) but untested here.

7.4 Statistical Conclusion Validity

Every confirmatory claim in this paper uses Holm-corrected paired inference across a prespecified family (Section 4.10), and the power analysis (Section 5.5) reports what that correction can and cannot detect: the study is well positioned to rule out a general effect at the scale of the observed point estimates (median absolute delta 0.0036), but effects below the Holm-adjusted minimum detectable scale (0.0201 at 80% power in the active larger-pool family) are not ruled out. A non-significant result under this design means the evidence does not support a general effect at a detectable scale; it does not establish exact equality between repaired and unrepaired rankings, and the narrow equivalence analysis (Section 5.5) found equivalence established for only a minority of cells at either tested margin, not the majority – this paper deliberately reports that as an open uncertainty rather than rounding it to a stronger equivalence claim. The real-LLM pilot's repeated per-query rows are handled with cluster-level ($n = 6$) inference specifically because the alternative, row-level treatment of the same data, is a documented, previously-identified source of overstated statistical confidence for this exact evidence (Section 4.10); six independent clusters remain a small basis for precise inference, and this is stated as a limitation of that subsection's evidence, not resolved by the cluster correction itself.

7.5 Computational Limitations

Exact minimum-weight feedback-arc-set solving reaches certified optimality quickly at the query-graph sizes studied here (mean solve time under ten milliseconds, Section 5.4), but this study does not evaluate exact solving at larger graph sizes and makes no claim about its scalability beyond the sizes tested. Its role in this paper is specifically diagnostic: a repair-method control used to rule out heuristic suboptimality as an explanation for a specific empirical finding (Section 6.3), not a claim that exact solving is a practical repair method for graphs materially larger than those in this study's candidate pools. Greedy cycle-peeling remains the practical choice for larger graphs precisely because exact minimum-weight feedback-arc-set solving is NP-hard in general (Section 2.3); this study's exact-repair results should be read as evidence about a specific, bounded regime, not as a general solved-problem claim.

8 Conclusion

This study separated four stages that are often collapsed in preference-graph ranking pipelines: graph construction, structural repair, ranking extraction, and retrieval evaluation. RQ1 asked how much structural inconsistency appears in derived multi-ranker preference graphs and how strongly it depends on construction choices. The answer is that construction is decisive: the conservative `ms2` regime produced acyclic graphs in this evidence, while the permissive `ms1` regime produced cyclic graphs for most queries, with much of the cyclicity coming from direct mutual contradiction. RQ2 asked what greedy and exact repair change structurally. Both methods restore acyclicity where there is structure to repair; exact SCIP-based repair removes systematically less edge weight than greedy repair on cyclic graphs, confirming that greedy over-removes relative to the true minimum-weight feedback arc set. RQ3 asked whether repair improves retrieval effectiveness. Across the canonical, larger-pool, and exact-repair comparison families, no repaired-versus-unrepaired nDCG comparison survives Holm correction, even though larger-pool repair can change which documents enter the evaluated top-k set. RQ4 asked whether the retrieval conclusion is robust to construction, pooling, extraction, cutoff, and repair optimality choices. The answer is yes within the study's scope: exact repair closes the heuristic-suboptimality explanation, but it does not reveal a retrieval gain hidden by greedy repair.

The practical implication is therefore not that acyclicity is irrelevant, or that repair can never help retrieval. It is that structural improvement and retrieval improvement are different claims requiring different evidence. Researchers using preference graphs should report construction choices, repair diagnostics, extraction rules, and downstream retrieval metrics separately, and should include an unrepaired baseline whenever repair is proposed as a task-improving intervention. Exact repair is especially useful as a bounded methodological control: it can test whether a weak or null downstream result is merely an artifact of heuristic repair, without being presented as a scalable production method.

Future work should test the same separation on broader large-language-model judgment benchmarks, additional graph-construction strategies, alternative ranking-extraction methods, and downstream objectives beyond nDCG-based retrieval. Such extensions would be able to identify settings where structural repair and task utility align more closely, while preserving the central discipline established here: acyclicity should be evaluated as a structural property first, and as a retrieval-quality intervention only when the downstream evidence supports that interpretation.

9 Data Availability and Reproducibility

The public benchmarks used in this study are available from their original sources: SciDocs, FiQA, HotpotQA, and BRIGHT are cited in Section 4.3. The code, fixed query lists, processed intermediates, figure data, and scripts required to reproduce the reported tables and figures are available at https://github.com/SoroushVahidi/consistency-aware-llm-rankin. The main-paper score-derived results are regenerated from stored ranker-score and relevance-judgment artifacts; no paid provider call is required to recompute those statistics. The bounded real-large-language-model pilot reported in Section 5.6 is supported by sanitized derived artifacts and usage summaries in the repository. Raw provider request and response payloads are not included in the repository, because they may contain prompt/completion content and operational details that are excluded by the project's artifact policy.

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Declarations

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Competing interests

The author declares no competing interests.

Ethics approval

Not applicable. This study uses only publicly available benchmark datasets and stored, derived score files; it involves no human subjects, no personal or identifiable data, and no new data collection from human participants.

Consent to participate

Not applicable; no human participants were involved in this study.

Consent for publication

Not applicable; no individual person's data are presented in this manuscript.

Data, materials, and code availability

The datasets used in this study are publicly available from their original sources and are cited in the manuscript. No new benchmark dataset was generated. The code and processed artifacts used to generate the reported results are available at https://github.com/SoroushVahidi/consistency-aware-llm-rankin. Raw provider request and response payloads from the bounded large-language-model pilot are not included in the repository for the reasons stated in Section 9.

Authors' contributions

This is single-author work. Soroush Vahidi conceived the study, designed and implemented the experimental pipeline, ran the experiments, analyzed and verified the results, prepared the figures and tables, and wrote the manuscript.

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